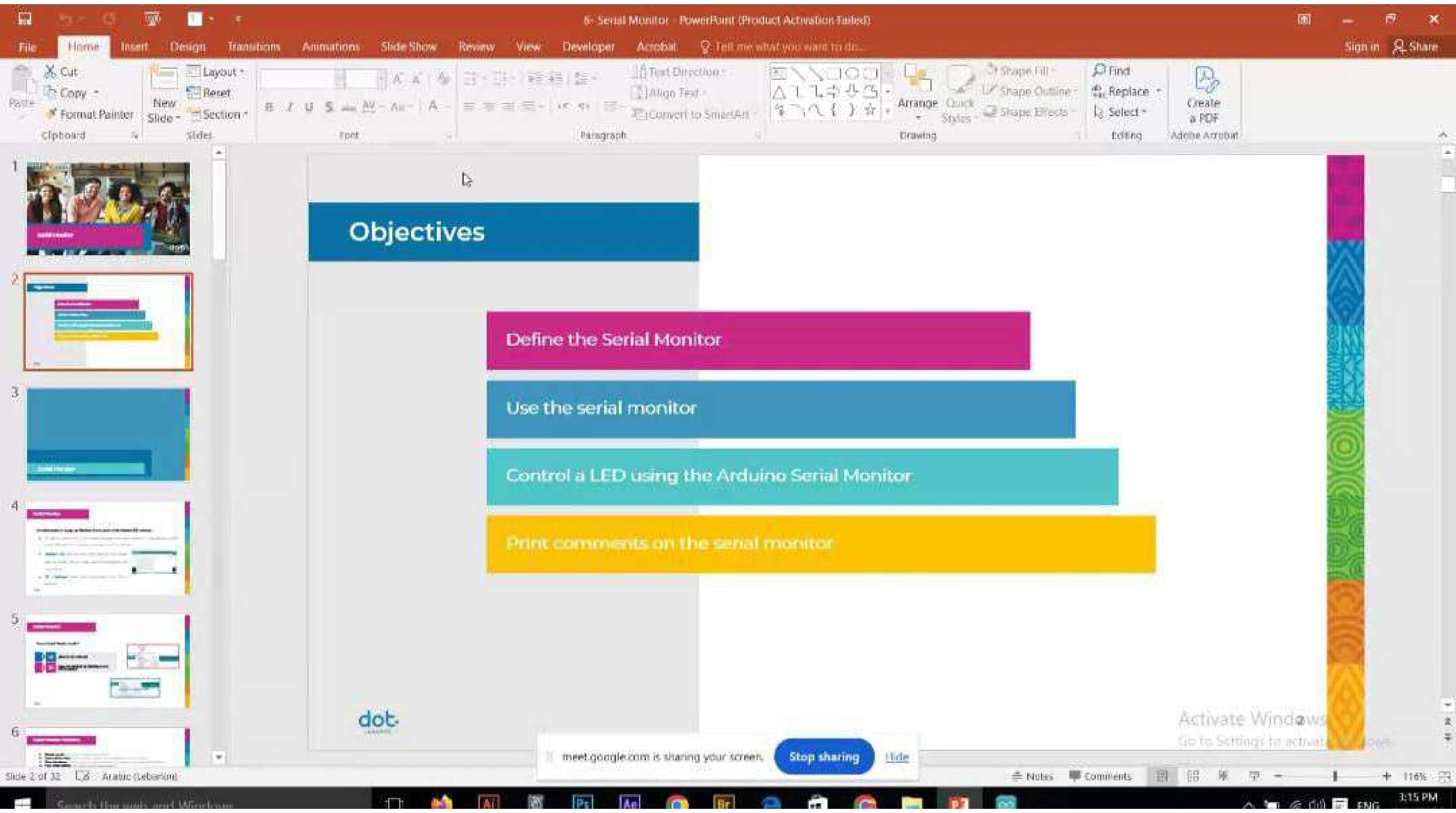
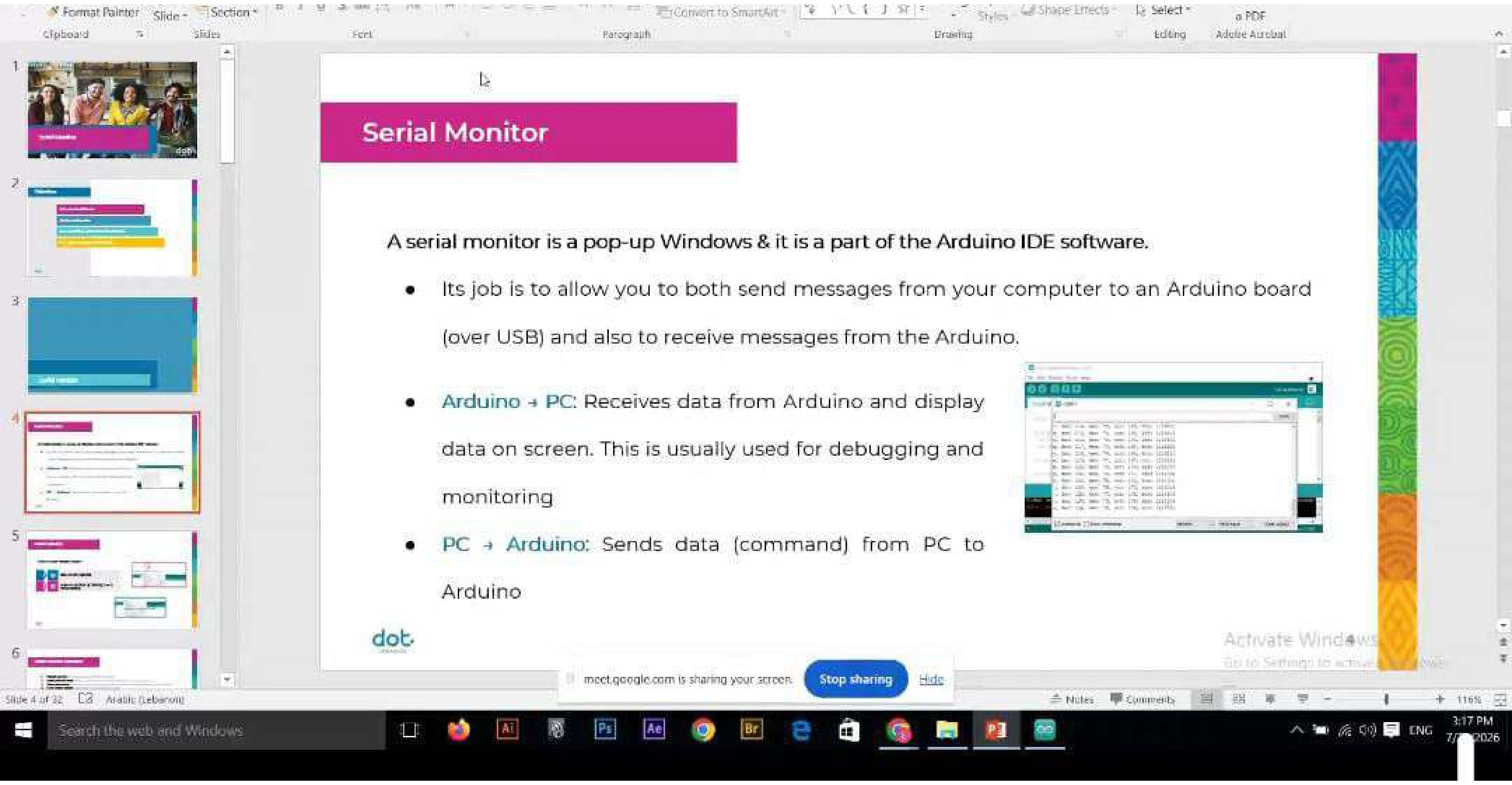

6- Serial Monitor - PowerPoint (Product Activation Failed)
Objectives
Define the Serial Monitor
Use the serial monitor
Control a LED using the Arduino Serial Monitor
Print comments on the serial monitor
dot.
meet.google.com is sharing your screen.
Stop sharing
Hide
Activate Windows
Slide 2 of 32


Serial Monitor
A serial monitor is a pop-up Windows & it is a part of the Arduino IDE software.
Its job is to allow you to both send messages from your computer to an Arduino board (over USB) and also to receive messages from the Arduino.
Arduino → PC: Receives data from Arduino and display data on screen. This is usually used for debugging and monitoring
PC → Arduino: Sends data (command) from PC to Arduino
dot
meet.google.com is sharing your screen.
Stop sharing
Hide
Activate Windows
Search the web and Windows
ENG
3:17 PM

Serial Monitor

A serial monitor is a pop-up Windows & it is a part of the Arduino IDE software.

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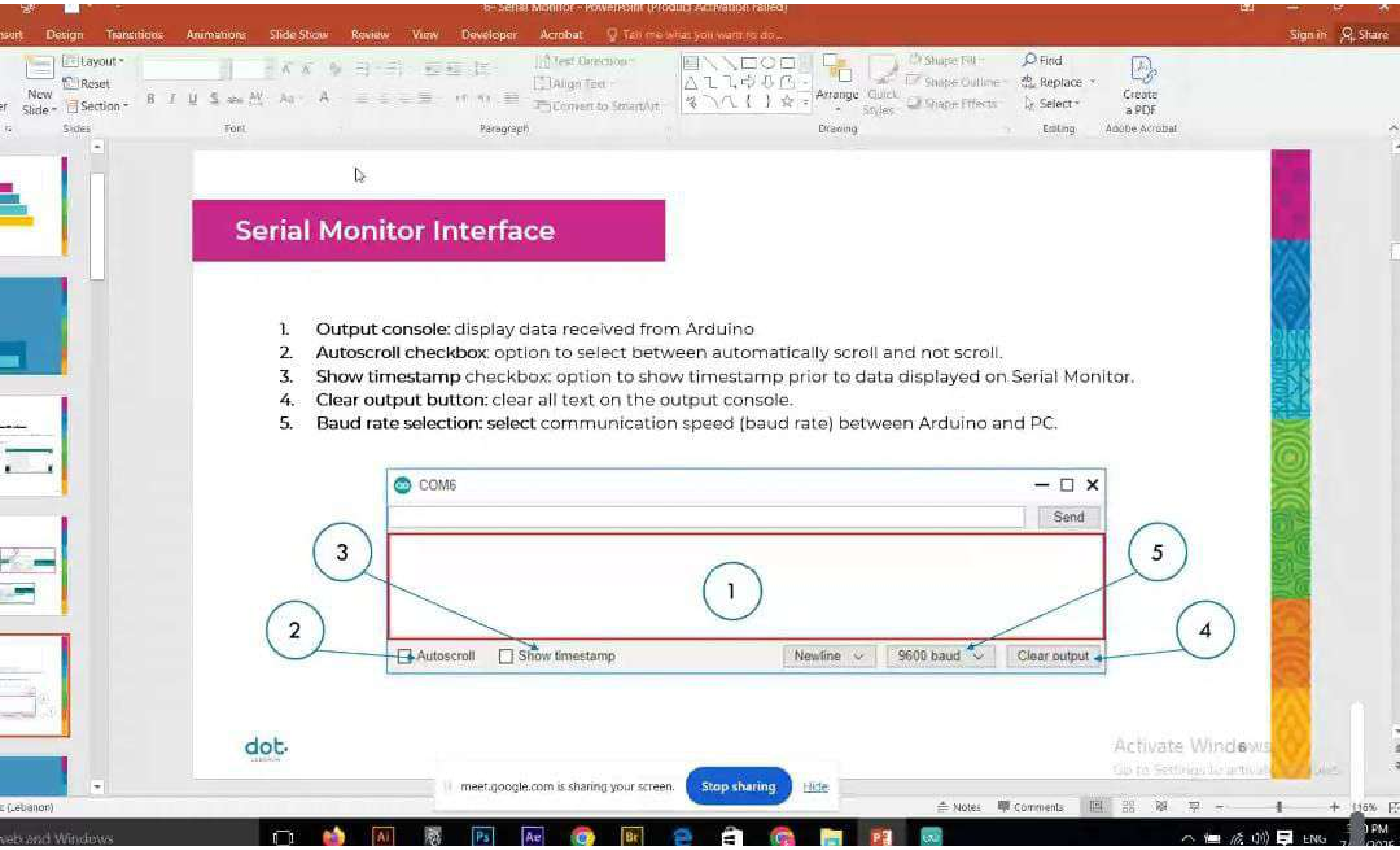
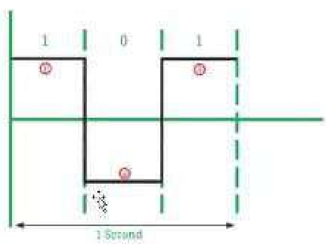
dot.


Serial Monitor Interface
1. Output console: display data received from Arduino
2. Autoscroll checkbox: option to select between automatically scroll and not scroll.
3. Show timestamp checkbox: option to show timestamp prior to data displayed on Serial Monitor.
4. Clear output button: clear all text on the output console.
5. Baud rate selection: select communication speed (baud rate) between Arduino and PC.
COM6
Send
1
2
3
4
5
Autoscroll
Show timestamp
Newline
9600 baud
Clear output
dot
meet.google.com is sharing your screen.
Stop sharing
Hide
Activate Windows

Serial Communication or Baud Rates

- A baud is a measure of how many times the hardware can send **0s** and **1s** in a second.
- The baud rate must be set properly for the board to convert incoming and outgoing information to useful data.
- The Arduino board can communicate at various baud ("baud rates").

Example:
If your receiver is expecting to communicate at a baud rate of 2400, but your transmitter is transmitting at a different rate (for example 9600), the data you get will not make sense.

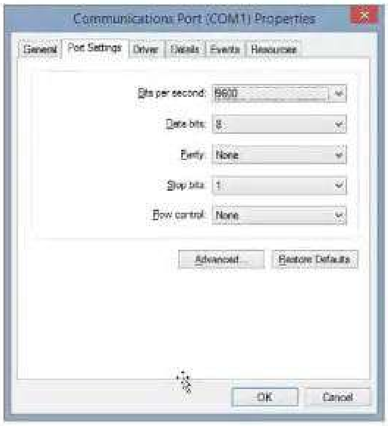


dot.

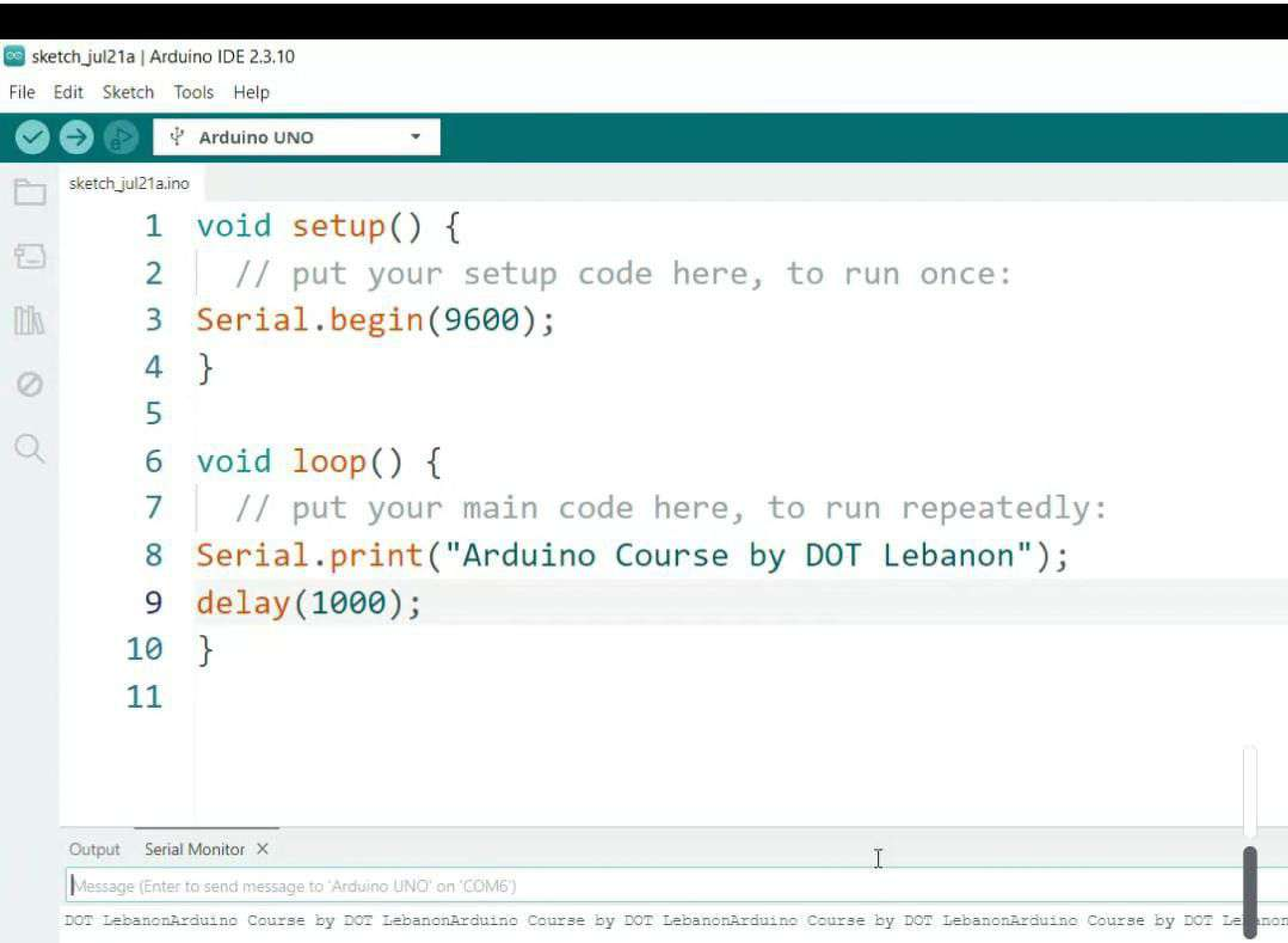
Serial Communication or Baud Rates

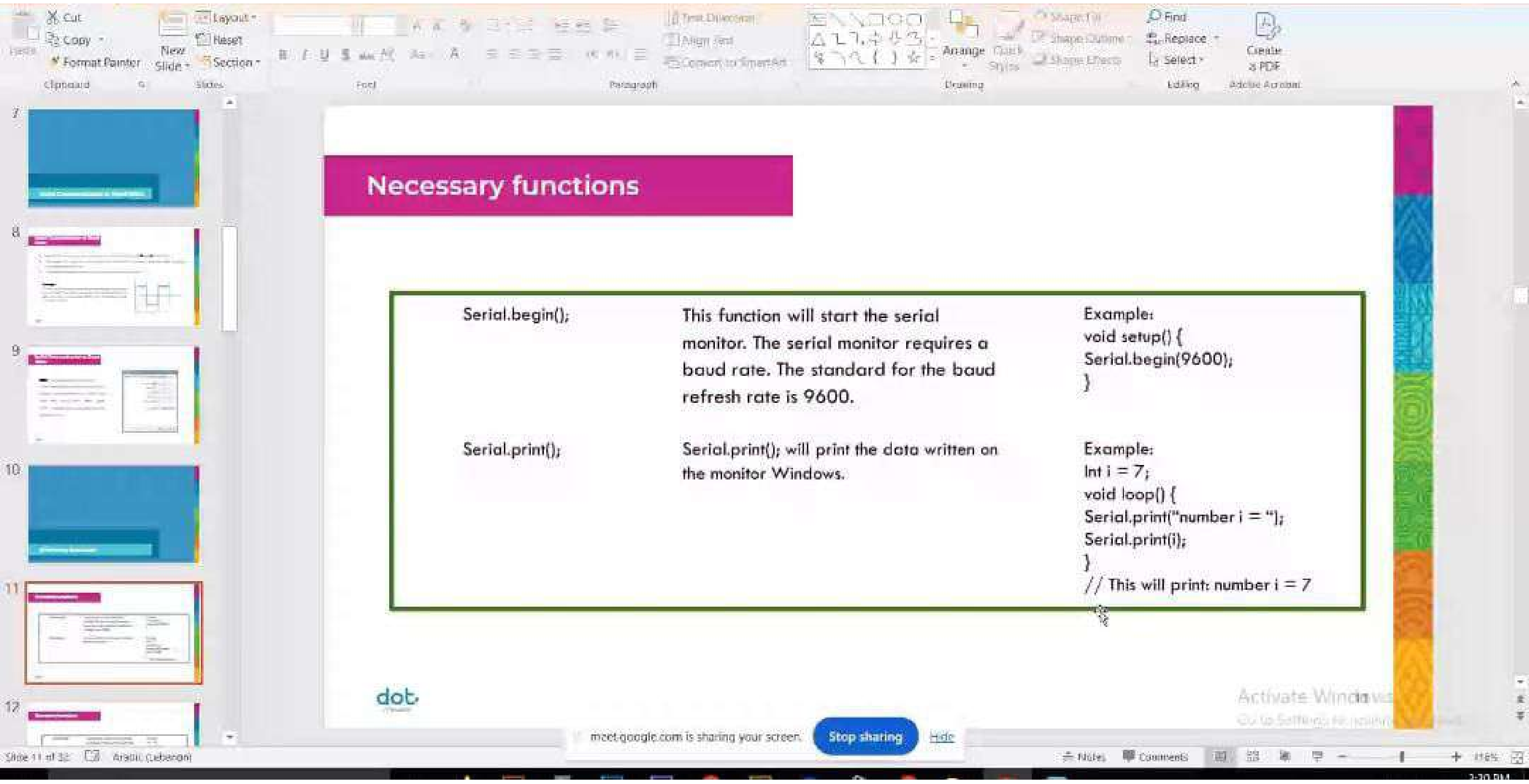
9600 is a good baud rate to start with.

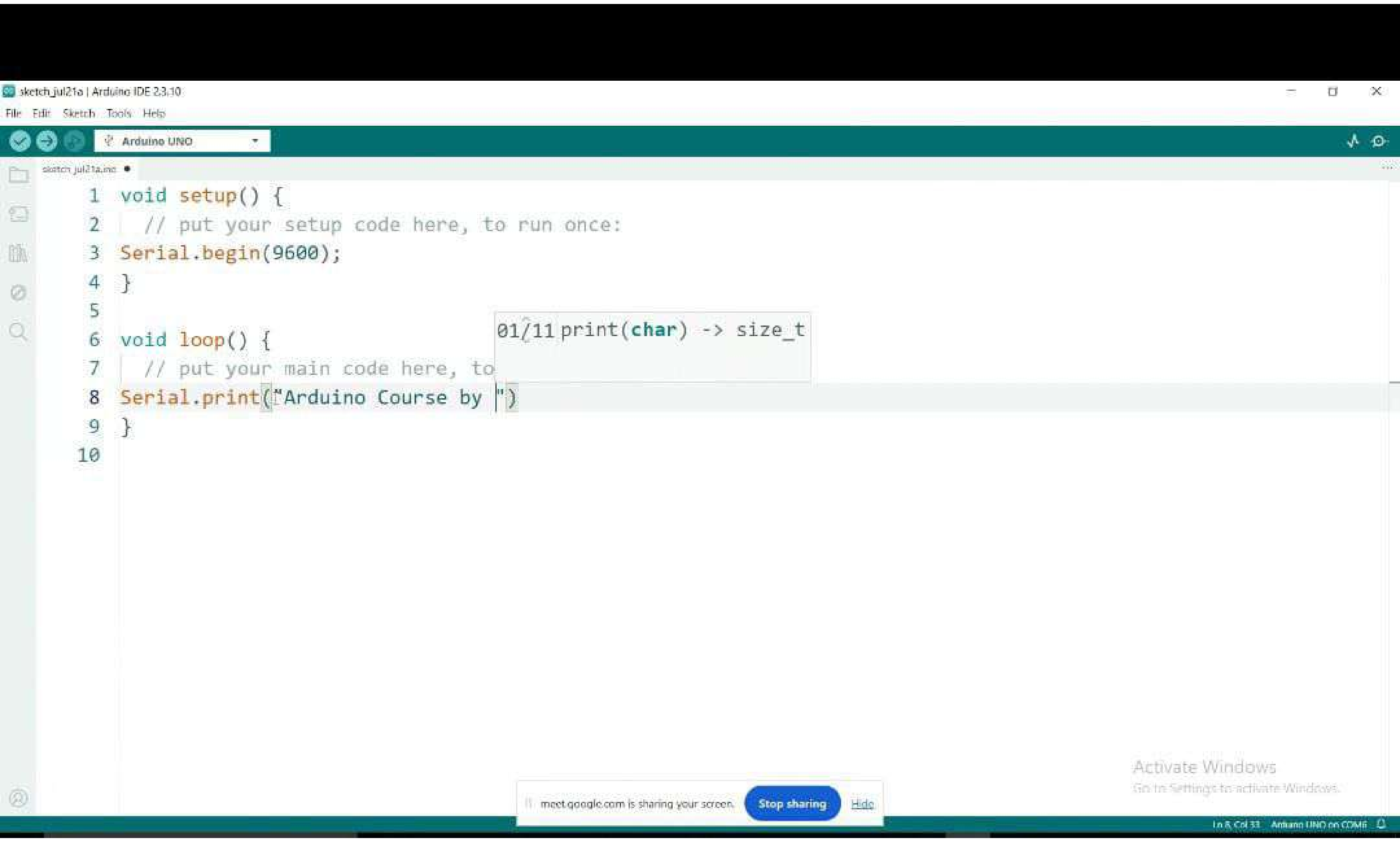
Other standard baud rates available on most Arduino modules include: 300, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600, or 115200 and you are free to specify other baud rates.

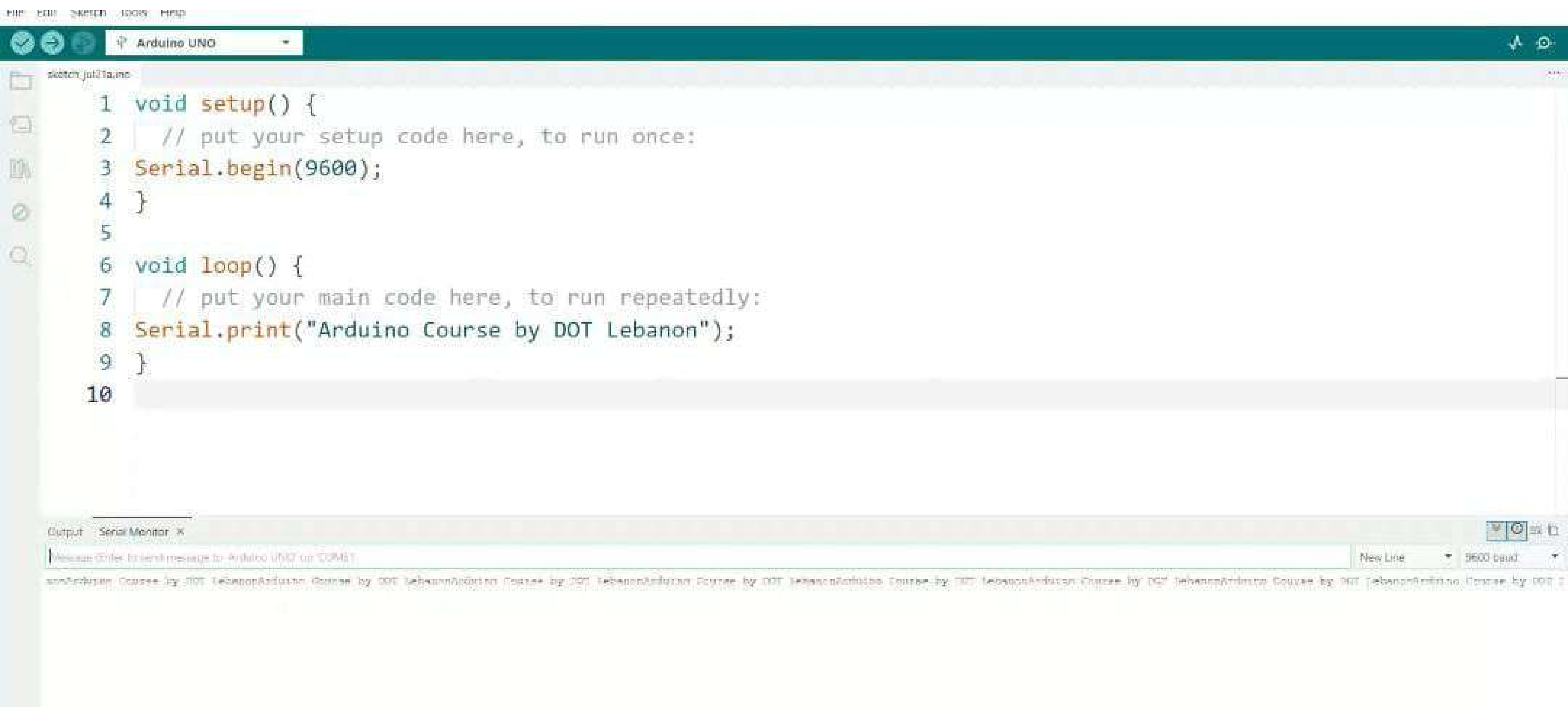


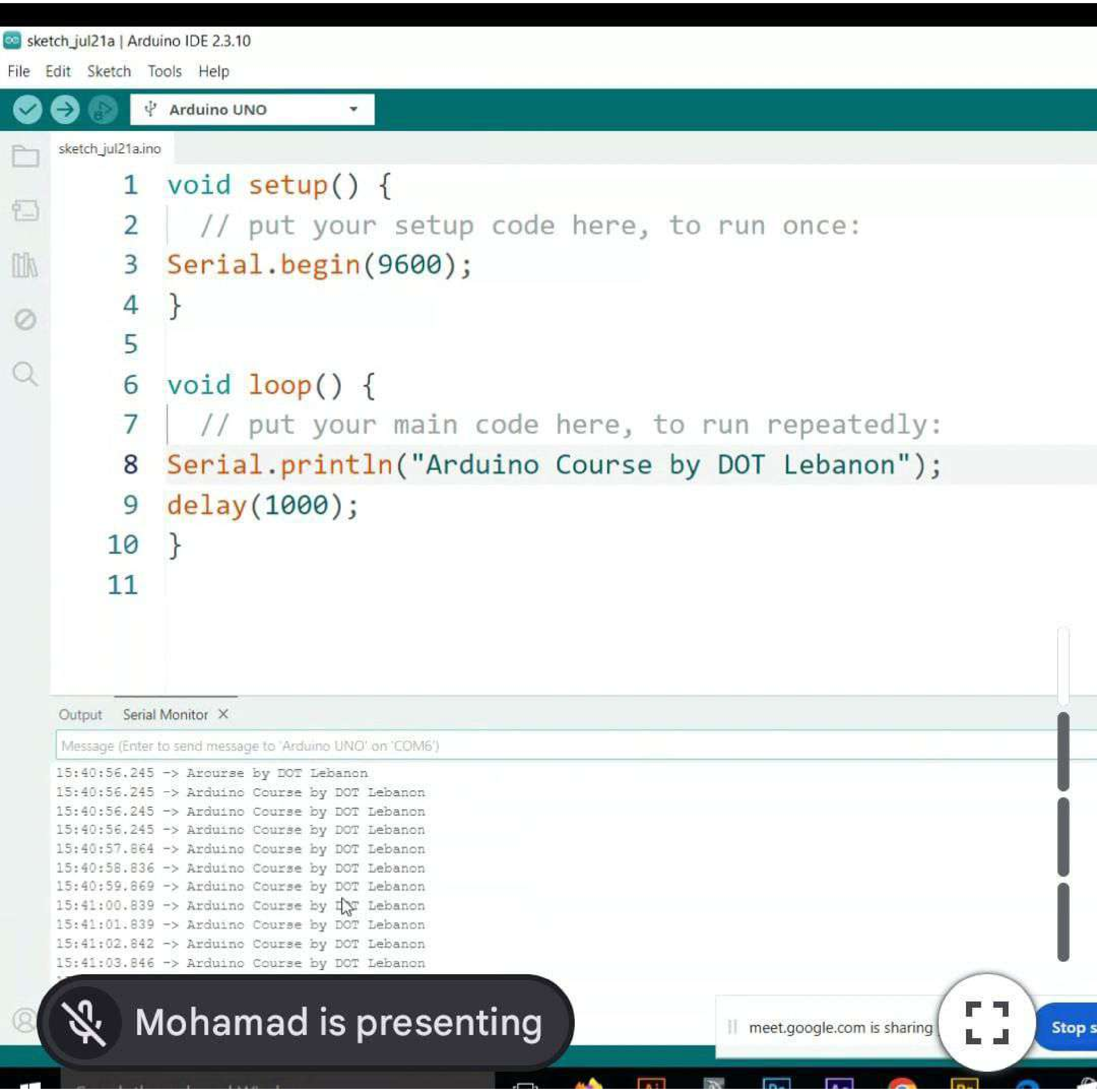
dot.

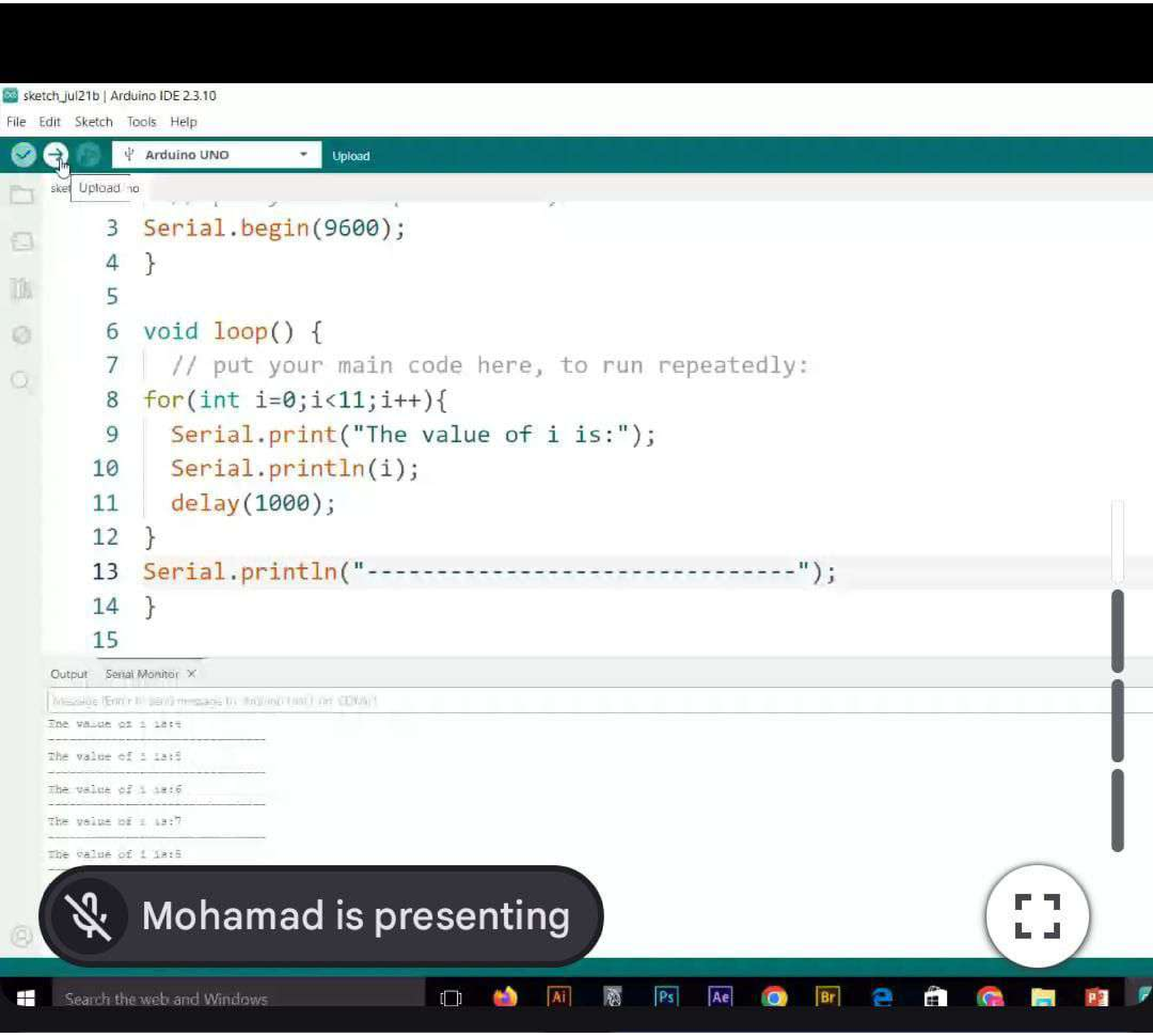

sketch_jul21a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul21a.ino
void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
Serial.print("Arduino Course by DOT Lebanon");
delay(1000);
}
Output
Serial Monitor
Message (Enter to send message to 'Arduino UNO' on 'COM6')
DOT LebanonArduino Course by DOT LebanonArduino Course by DOT LebanonArduino Course by DOT LebanonArduino Course by DOT Le


Necessary functions
Serial.begin();
This function will start the serial monitor. The serial monitor requires a baud rate. The standard for the baud refresh rate is 9600.
Example:
void setup() {
Serial.begin(9600);
}
Serial.print();
Serial.print(); will print the data written on the monitor Windows.
Example:
Int i = 7;
void loop() {
Serial.print("number i = ");
Serial.print(i);
}
// This will print: number i = 7
dot
meet.google.com is sharing your screen.
Stop sharing
Hide

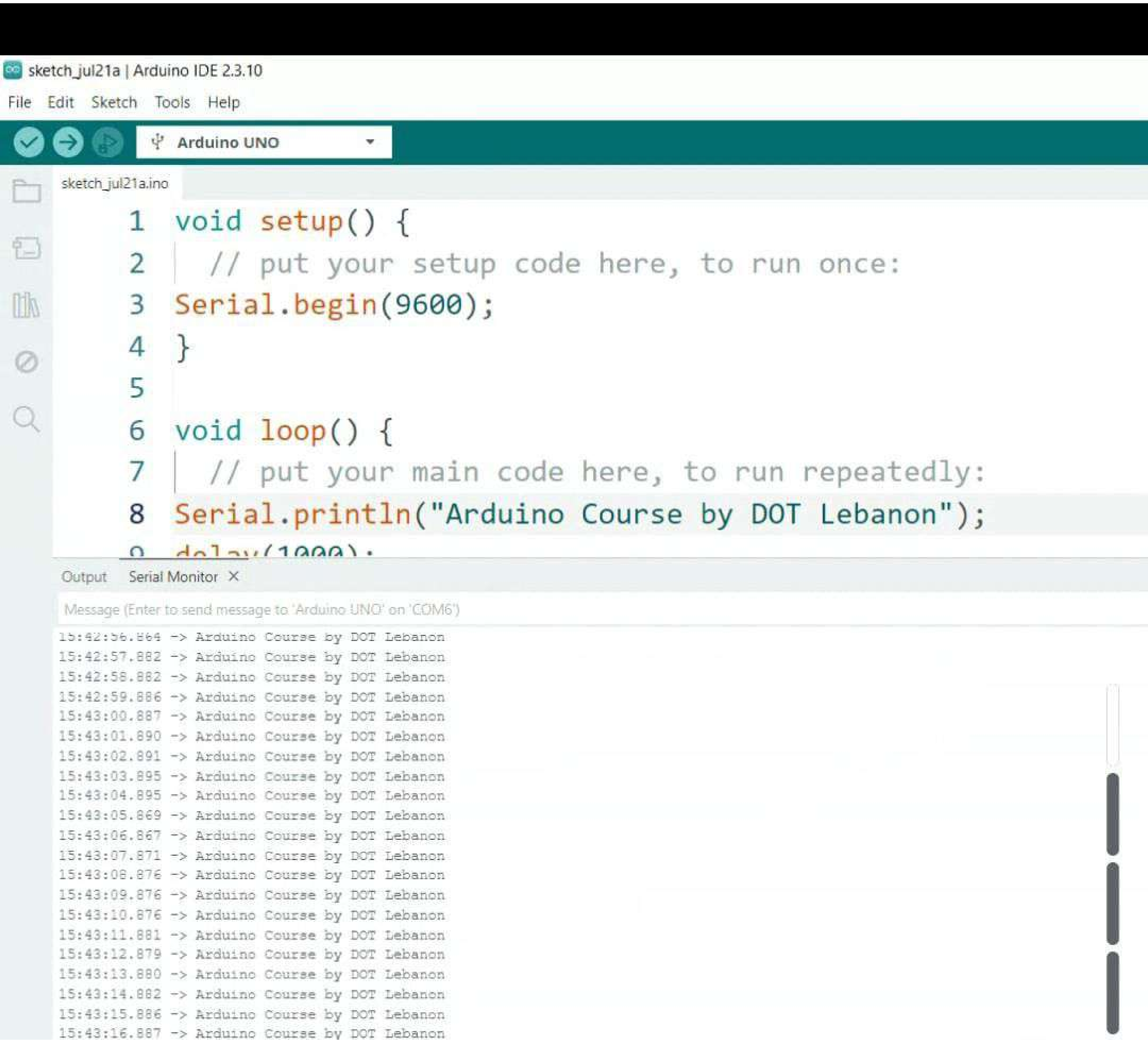

sketch_jul21a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul21a.ino
void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
}
void loop() {
// put your main code here, to
01/11 print(char) -> size_t
Serial.print("Arduino Course by ")
}
Activate Windows
meet.google.com is sharing your screen.
Stop sharing
Hide

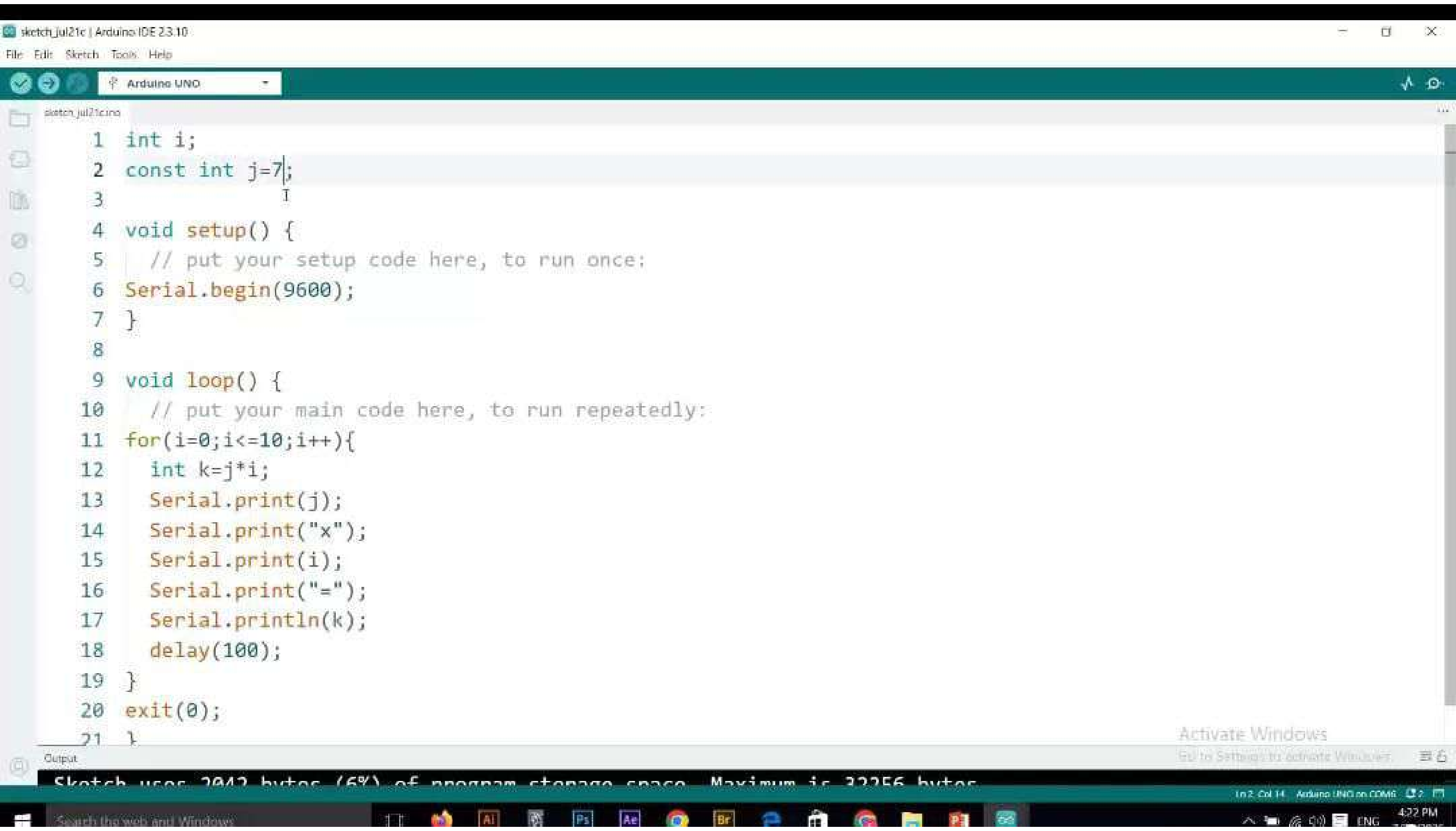

Arduino UNO
void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
Serial.print("Arduino Course by DOT Lebanon");
}
Output
Serial Monitor
New Line
9600 baud

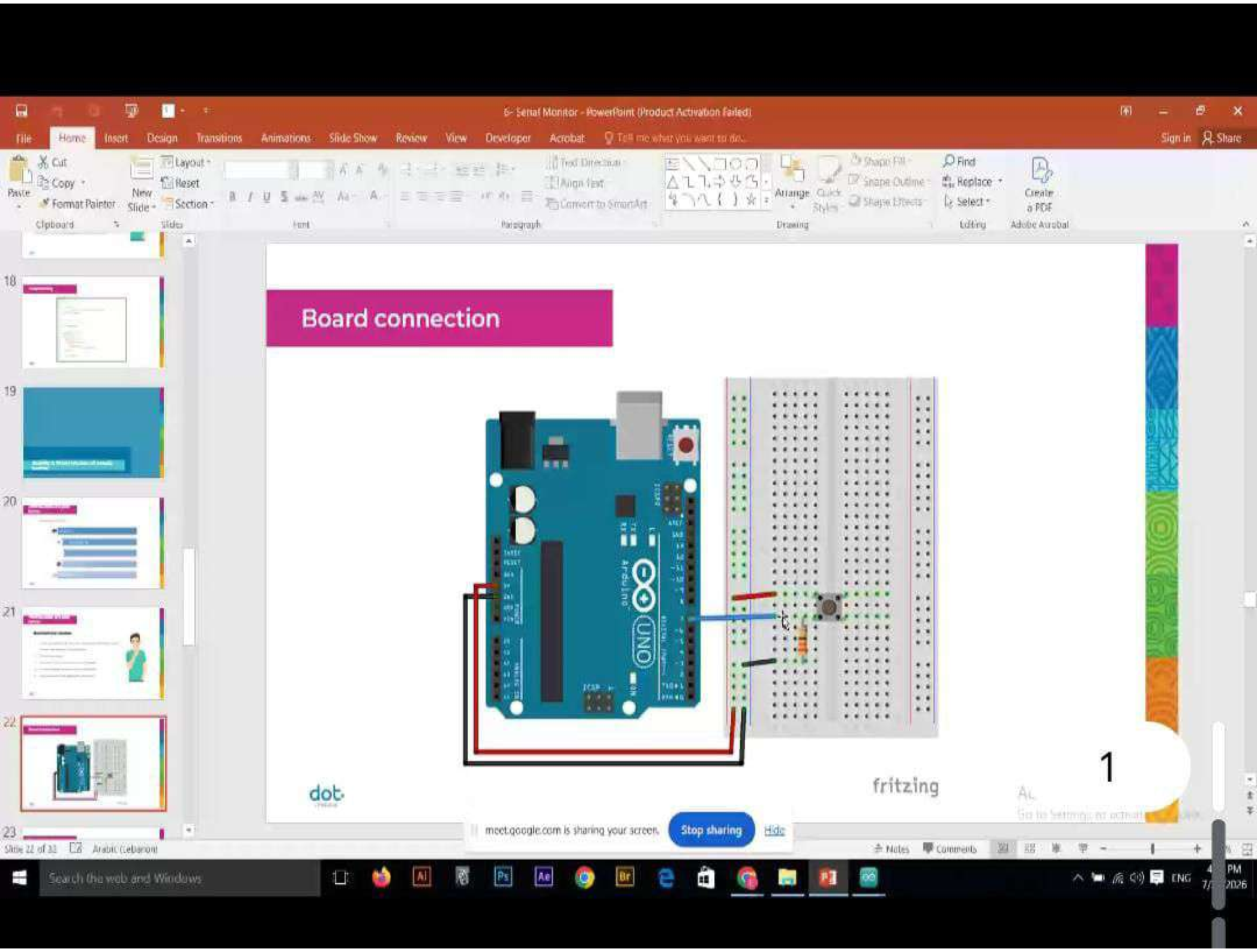

sketch_jul21a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul21a.ino
void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
Serial.println("Arduino Course by DOT Lebanon");
delay(1000);
}
Output
Serial Monitor
Message (Enter to send message to 'Arduino UNO' on 'COM6')
15:40:56.245 -> Arourse by DOT Lebanon
15:40:56.245 -> Arduino Course by DOT Lebanon
15:40:56.245 -> Arduino Course by DOT Lebanon
15:40:56.245 -> Arduino Course by DOT Lebanon
15:40:57.864 -> Arduino Course by DOT Lebanon
15:40:58.836 -> Arduino Course by DOT Lebanon
15:40:59.869 -> Arduino Course by DOT Lebanon
15:41:00.839 -> Arduino Course by
15:41:01.839 -> Arduino Course by DOT Lebanon
15:41:02.842 -> Arduino Course by DOT Lebanon
15:41:03.846 -> Arduino Course by DOT Lebanon
Mohamad is presenting
meet.google.com is sharing

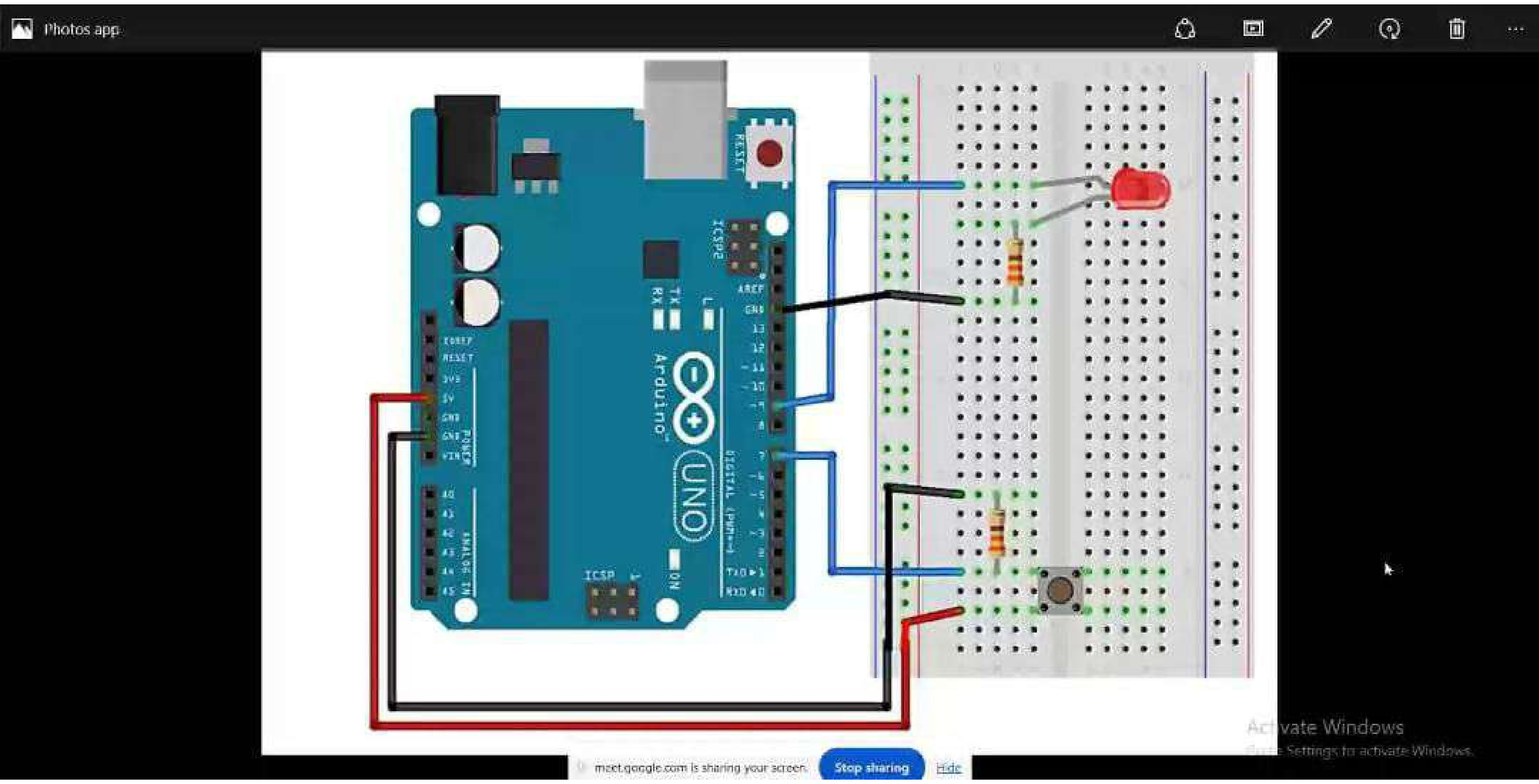


sketch_jul21b | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
Upload
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
for(int i=0;i<11;i++){
Serial.print("The value of i is:");
Serial.println(i);
delay(1000);
}
Serial.println("----------------------------");
}
Output
Serial Monitor
The value of i is:6
The value of i is:7
Mohamad is presenting
Search the web and Windows


sketch_jul21a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul21a.ino
void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
Serial.println("Arduino Course by DOT Lebanon");
Output
Serial Monitor
Message (Enter to send message to 'Arduino UNO' on 'COM6')
15:42:56.864 -> Arduino Course by DOT Lebanon
15:42:57.882 -> Arduino Course by DOT Lebanon
15:42:58.882 -> Arduino Course by DOT Lebanon
15:42:59.886 -> Arduino Course by DOT Lebanon
15:43:00.887 -> Arduino Course by DOT Lebanon
15:43:01.890 -> Arduino Course by DOT Lebanon
15:43:02.891 -> Arduino Course by DOT Lebanon
15:43:03.895 -> Arduino Course by DOT Lebanon
15:43:04.895 -> Arduino Course by DOT Lebanon
15:43:05.869 -> Arduino Course by DOT Lebanon
15:43:06.867 -> Arduino Course by DOT Lebanon
15:43:07.871 -> Arduino Course by DOT Lebanon
15:43:08.876 -> Arduino Course by DOT Lebanon
15:43:09.876 -> Arduino Course by DOT Lebanon
15:43:10.876 -> Arduino Course by DOT Lebanon
15:43:11.881 -> Arduino Course by DOT Lebanon
15:43:12.879 -> Arduino Course by DOT Lebanon
15:43:13.880 -> Arduino Course by DOT Lebanon
15:43:14.882 -> Arduino Course by DOT Lebanon
15:43:15.886 -> Arduino Course by DOT Lebanon
15:43:16.887 -> Arduino Course by DOT Lebanon


sketch_jul21c | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul21c.ino
int i;
const int j=7;
void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
for(i=0;i<=10;i++){
int k=j*i;
Serial.print(j);
Serial.print("x");
Serial.print(i);
Serial.print("=");
Serial.println(k);
delay(100);
}
exit(0);
}
Activate Windows
Output
Sketch uses 2042 bytes (6%) of program storage space. Maximum is 32256 bytes.
Search the web and Windows
ENG
4:22 PM


6- Serial Monitor - PowerPoint (Product Activation Failed)
File
Home
Insert
Design
Transitions
Animations
Slide Show
Review
View
Developer
Acrobat
Sign in
Share
Board connection
dot
fritzing
1
meet.google.com is sharing your screen.
Stop sharing
Hide
Search the web and Windows
ENG


Photos app
Arduino
UNO
meet.google.com is sharing your screen.
Stop sharing
Hide

Arduino UNO

sketch_jul21e.ino

```
  // put your setup code here, to run once:
pinMode(led,OUTPUT);
pinMode(button,INPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
ButtonState=digitalRead(button);

if(ButtonState==HIGH){
  digitalWrite(led,HIGH);
  Serial.println("The button is PRESSED");
  delay(500);
}
else{
  digitalWrite(led,LOW);
  Serial.println("The buttons is NOT PRESSED");
  delay(500);
}
```

Output Serial Monitor

sketch_jul21f | Arduino IDE 2.3.10

```
int button=7;
int ButtonState;

void setup() {
  // put your setup code here, to run once:
pinMode(button,INPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
ButtonState=digitalRead(button);
Serial.print("The button is :");
Serial.println((ButtonState==HIGH)?"PRESSED":"NOT PRESSED");
delay(500);
}
```